

Design and conduct repeated chance experiments and simulations using digital tools to model conditional probability and interpret results

Learning intention

- design and conduct repeated chance experiments and simulations using digital tools to model conditional probability and interpret results.

Teach and model

- using simulation to predict the number of people likely to be infected with a strain of flu or virus.
- using simulations to gather data on frequencies for situations involving chance that appear to be counter-intuitive.
- recognising that an event can be dependent on another event and that this will affect the way its probability is calculated.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.